

# Assignments

Topic-wise practice worksheets, one folder per week of the plan in `../docs/03-practice-plan.md`. Each folder (once built) contains:

- `assignment.md` — the questions (4-option MCQ, Class 4 difficulty, mixing direct computation with word-problem/Everyday-Maths style framing)
- `answer-key.md` — correct options with a short worked explanation each

## Folder list (per the 12-week plan)

| Folder                          | Topic                                                                                                | Status                  |
|---------------------------------|------------------------------------------------------------------------------------------------------|-------------------------|
| 01-number-sense/                | Place value, number names, ordering, rounding, Roman numerals                                        | ✓ Built                 |
| 02-four-operations/             | Addition, subtraction, multiplication, division (multi-digit + word problems)                        | ✓ Built                 |
| 03-factors-multiples-patterns/  | Factors, multiples, number patterns                                                                  | ✓ Built                 |
| 04-fractions-decimals/          | Comparing & equivalent fractions, simple decimals                                                    | ✓ Built                 |
| 05-measurement-money/           | Length, weight, capacity, money, conversions                                                         | ✓ Built                 |
| 06-time-calendar/               | Clock time, duration, calendar                                                                       | ✓ Built                 |
| 07-geometry-perimeter-area/     | Shapes, symmetry, perimeter, basic area                                                              | ✓ Built                 |
| 08-data-handling/               | Tally marks, pictographs, bar graphs                                                                 | ✓ Built                 |
| 09-reasoning-patterns-coding/   | Number/figure patterns, analogy, classification, alphabet test, coding–decoding                      | ✓ Built                 |
| 10-reasoning-direction-figures/ | Ranking, direction sense, days & dates, mirror images, embedded figures, geometrical solids          | ✓ Built (24 Q)          |
| 11-achievers-hots/              | Multi-step HOTS problems across all topics above                                                     | ✓ Built                 |
| 12-math-kangaroo-stretch/       | <b>Optional</b> — Math Kangaroo-style puzzles (shape counting, mirror images, logic, cube nets)      | ✓ Built (15 Q, 3 tiers) |
| 13-achievers-level2/            | <b>Level-2 track</b> — harder, multi-step HOTS beyond Week 11, for students who've qualified Level 1 | ✓ Built                 |

| Folder               | Topic                                                                                                | Status  |
|----------------------|------------------------------------------------------------------------------------------------------|---------|
| 14-reasoning-level2/ | <b>Level-2 track</b> — compound logical-reasoning puzzles stacking 2+ clues/rules, beyond Weeks 9–10 | ✓ Built |
| 15-geometry-level2/  | <b>Level-2 track</b> — compound-shape perimeter & area (paths, frames, cut corners), beyond Week 7   | ✓ Built |

Folders `01` – `11` are the core 12-week plan, built in the same two-file ( `assignment.md` + `answer-key.md` ) shape, 15 questions each — except `10-reasoning-direction-figures/` , which has 24 (15 text-based + 9 diagram-based). `12-math-kangaroo-stretch/` is the optional extension described in `docs/03-practice-plan.md` , not part of the core plan or any Indian olympiad's syllabus. `08-data-handling/` , `10-reasoning-direction-figures/` , and `12-math-kangaroo-stretch/` each ship an `images/` folder of SVG diagrams referenced from their `assignment.md` .

Folders `13` – `15` are a **Level-2 topic track**: for students who've qualified Level 1 of the real exam and want harder, topic-focused practice rather than just the mixed-topic Level-2 sample papers ( `sample-papers/sample-paper-06.md` onward). Same two-file shape, 15 questions each, every answer verified programmatically. These three topics were picked because they're where Level 2 gets *conceptually* harder, not just numerically bigger — the other core topics (number sense, four operations, data handling, etc.) are already covered by the harder numbers in the Level-2 sample papers.

## Diagram-based question types

Mirror images, embedded figures, and geometrical-solid recognition are inherently visual Section-1 (Logical Reasoning) sub-topics. Rather than describing them in text, `10-reasoning-direction-figures/images/` has an SVG diagram for each question:

- **Mirror images** — the original figure plus 4 candidate reflections (one correctly flipped left-right, the other three are unchanged / flipped upside-down / rotated 180° — classic distractors). The correct option is generated by an exact `scale(-1,1)` transform, not hand-drawn, so it's guaranteed accurate.
- **Embedded figures** — a composite figure made of two overlapping shapes; find which of 4 candidate shapes is traceable inside it.
- **Geometrical solids** — a drawn cube, cone, and cylinder, each with a question about its name or face count.

`08-data-handling/images/` similarly has a real SVG pictograph and bar graph (rather than an emoji/markdown-table stand-in) for its data-handling questions.

[12-math-kangaroo-stretch/images/](#) has 7 more SVGs for its puzzle questions — shape-counting figures, a mirror-image puzzle (same guaranteed-correct transform technique as above), a balance-scale illustration, a grid for path-counting, a shaded-fraction figure, and a labelled cube net.